

Artificial Intelligence as a Learning Partner in Medication Calculation Education: A Pedagogical Framework Based on Flipped Classroom, Prompt Engineering, and Reflective Portfolios

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Abstract—Medication calculation is a high-stakes competence in nursing and health education, where errors can have fatal consequences. The rapid integration of generative AI into higher education creates both opportunities and risks for how this competence is taught and assessed. This paper is the result of a cooperation between the Faculty of Technology, Environmental and Social Sciences and the Faculty of Health and Social Sciences at Western Norway University of Applied Sciences (HVL). Based on classroom experiences, and informed by recent literature, we present a pedagogical framework that integrates three core elements: (1) prompt engineering as a prerequisite for active engagement with AI, (2) flipped classroom design centred on inquiry-based medication calculation tasks, and (3) a structured *learning portfolio* (læringsmappe) combining reflection, self-assessment, and algorithmic thinking. We identify six major pitfalls of uncritical AI use in medication calculation and propose pedagogical countermeasures. A concrete case study (dose calculation, IV drip rate, and unit conversion) illustrates how AI can be integrated as a reflective guide rather than a solution provider. Preliminary findings indicate that students who engage in structured reflection and self-assessment develop greater metacognitive awareness and are less likely to rely passively on AI-generated answers. The paper contributes a replicable instructional design and an exploratory evaluation of its effects on student learning strategies, clinical reasoning, and professional accountability.

Index Terms—Artificial intelligence, medication calculation, nursing education, prompt engineering, flipped classroom, reflective learning, algorithmic thinking, metacognition.

I. INTRODUCTION

In just a few years, artificial intelligence has moved from research laboratories into the daily workflow of students and lecturers. At the Faculty of Technology, Environmental and Social Sciences and the Faculty of Health and Social Sciences at Western Norway University of Applied Sciences (HVL), we have observed that AI tools such as ChatGPT are now

used by the majority of students for proofreading, structuring arguments, solving mathematical problems, and even generating code. This shift has happened faster than our educational systems could adapt [1].

However, attendance in exercise sessions has declined, and students increasingly rely on AI for guidance rather than engaging with teachers or peers. This is particularly concerning in medication calculation, where precision and clinical judgment are essential and where errors can lead to patient harm [6], [7]. To counter this, we must design pedagogical activities that teach students how to use AI as a reflective tool, and that help them become aware of their own learning processes.

For lecturers, AI is transforming our opportunities while also presenting new challenges. On one hand, we can now generate personalised problem sets, analyse student misconceptions at scale, and design flipped classroom activities that were previously too time-consuming. On the other hand, traditional take-home assignments are no longer viable because AI can solve them instantly. Recent research shows that GenAI can attain performance comparable to first-class degree level in first-year undergraduate mathematics examinations, highlighting the urgent need for assessment redesign [8].

Today's students differ from previous cohorts. They often have less theoretical background, struggle with abstract thinking, and are accustomed to learning through problem-solving and algorithmic thinking rather than theoretical exposition. Traditional one-way communication in lectures is becoming outdated. As a result, the lecturer's role is changing from a transmitter of knowledge to a facilitator and guide. This paper presents a pedagogical approach for helping students take ownership of their learning, using structured reflection and self-assessment [2], [3].

The interdisciplinary cooperation between the two faculties is central to this work. Medication calculation sits at the intersection of mathematics, technology, and clinical practice, and a meaningful pedagogical framework must integrate all three perspectives. The Faculty of Technology, Environmental and Social Sciences contributes expertise in mathematics, algorithmic thinking, and AI literacy, while the Faculty of Health and Social Sciences contributes clinical expertise in medication safety, nursing practice, and patient-centred care. Together, these perspectives form the foundation for the framework presented in this paper.

This paper proposes and evaluates a pedagogical framework for AI-supported medication calculation learning that integrates three core elements: (1) prompt engineering as a prerequisite for active engagement, (2) flipped classroom design centred on inquiry-based tasks, and (3) a structured learning portfolio that combines reflection, self-assessment, and algorithmic thinking.

The contribution of this paper is twofold. First, we provide a concrete and replicable instructional design that operationalises the role of generative AI as a learning partner rather than a solution provider in a high-stakes domain. Second, we present an exploratory evaluation of this framework based on classroom implementation, highlighting how structured reflection and guided AI use influence student learning processes, clinical reasoning, and metacognitive awareness.

The study is guided by the following research question: *How does a structured pedagogical framework for AI-supported learning influence students' engagement, learning strategies, and metacognitive awareness in medication calculation?*

II. RELATED WORK: AI IN HEALTH AND MATHEMATICS EDUCATION

Recent research on AI in education is organised around two main perspectives: the teacher's and the student's.

A. Teacher Perspectives: AI as a Tool for Designing Learning

From the teacher's side, generative AI can enrich differentiated instruction and support professional development [9]. AI can generate practice problems, quizzes, and complete assignments tailored to different ability levels. Cho and Kim [10] analysed AI-based platforms (KnowRe Math, ALEKS) and found that two-way adaptation works best, but none offered flexibility in presentation format, a gap teachers can fill manually.

Holman et al. [11] conducted a systematic review of 14 empirical studies (post-2020) on AI interventions in mathematics education. They found that intelligent tutoring systems (ITS) significantly improve learning outcomes when used over longer periods, and tools like Photomath showed a large effect size ($d = 1.37$) in pre-calculus. However, shorter interventions or those without substantial teacher involvement yielded smaller effects. The KI@school project [9] explicitly focuses on how generative AI can foster mathematical thinking and reflective practice.

In health education, similar trends are observed. A systematic review on GenAI and personalisation [13] confirms that the effectiveness of such systems depends on teacher-AI collaboration, data privacy, and equitable access. Ethical and regulatory challenges are further discussed in [14].

B. Student Perspectives: Learning to Use AI as a Learning Partner

From the student's side, the main challenge is to use AI as a training partner rather than a substitute. Zhang and Yang [15] emphasised that without structured teacher guidance on prompting, students risk over-reliance and shallow learning. Nikolic et al. [16] noted that while GenAI shows promise as a supplementary tool, minor inaccuracies are frequently observed. The literature has identified several risks associated with uncritical AI use, including over-dependence, loss of mathematical identity, and the development of a "techno-authoritative conviction" scheme [17]–[19]. Several studies have also documented AI hallucinations as a persistent problem [20]–[22]. Wulff and Kubsch [23] discuss the double-edged sword of GenAI in STEM education.

In medication calculation specifically, the consequences of errors are more severe than in general mathematics. The World Health Organization [7] identifies medication errors as a leading cause of preventable patient harm, and Reason's [6] work on human error emphasises that systems must be designed to make errors less likely and less harmful. This makes the pedagogical design of AI-supported medication calculation particularly important.

III. METHOD

This study follows an exploratory case study design situated in authentic classroom practice with a cohort of nursing and engineering students ($N=145$) at HVL. The aim is not to provide statistically generalisable results, but to investigate how a structured pedagogical framework for AI-supported learning functions in practice and how students interact with AI when guided by explicit instructional design. Data sources include student-produced prompts, written reflections, learning portfolios, and observational insights from classroom implementation.

The data used in this exploratory study consist primarily of qualitative sources, including student-generated prompts, written reflections, and observational notes from classroom interactions. These data were analysed through an interpretive approach, focusing on identifying recurring patterns in how students engaged with AI and reflected on their own learning processes.

The analysis was conducted using a thematic coding approach, where recurring patterns in student prompts and reflections were identified and categorised. The coding process was iterative and involved multiple passes through the data to refine emerging themes, informed by concepts from self-regulated learning theory [3] and constructive alignment [2].

Our pedagogical approach integrates three components: (1) explicit prompt engineering training, (2) a flipped-classroom investigation using medication calculation tasks,

and (3) a structured learning portfolio with reflection and self-assessment.

A. Prompt Engineering as a Precondition for Active Use

Students were taught that good prompts elicit guidance, not just answers. This is particularly important in medication calculation, where an incorrect answer can have clinical consequences.

“How do you calculate medicine?”

This prompt is too vague. The AI does not know which medication, which patient group, which route of administration, or which type of calculation is intended.

“I need to administer Furosemide 10 mg IV. The ampoule is 40 mg/ml. Can you show me step by step how to calculate the volume (ml) I need to draw up, and explain which formula you use?”

This prompt is concrete: it specifies the drug, strength, desired dose, available concentration, and requests both calculation and explanation. It invites a dialogue-based verification rather than answer-seeking [15].

B. Flipped Classroom Case Study

Students were given an assignment to solve medication calculation problems without prior instruction on the specific methods. The assignment was titled “*Flipped classroom with AI as a learning partner*”. However, they received a short introduction to the assignment itself, learning how to use AI as a successful advisor. The problem was to:

- 1) Explain the basic dose calculation formula and the IV drip rate formula.
- 2) Develop step-by-step solutions for tablet dosages, weight-based dosages, and IV infusions.
- 3) Reflect on strengths and weaknesses of each approach.

Students were also given structured reflection questions to guide their post-task analysis. Students documented their work in a learning portfolio, then wrote a short reflection.

From a theoretical perspective, this flipped classroom design aligns with constructivist and inquiry-based learning principles, where students actively construct understanding through exploration rather than passive reception [2]. The integration of AI as a dialogic partner extends this model by enabling iterative explanation, immediate feedback, and adaptive support during the learning process.

Recent research suggests that combining flipped classroom approaches with AI tools can enhance student engagement and self-regulated learning, particularly when students are required to actively articulate their reasoning and reflect on feedback [24]. This is consistent with findings from intelligent tutoring

systems research, where sustained interaction and feedback are critical for learning gains [11].

However, unlike traditional ITS, generative AI introduces epistemic uncertainty, requiring students to critically evaluate responses rather than accept them as authoritative. As emphasised by Zhang and Yang [15], explicit scaffolding in prompting and evaluation is therefore essential. In this context, the flipped classroom does not merely shift content delivery outside the classroom, but restructures the cognitive processes through which students engage with both medication calculation and AI.

IV. THE LEARNING PORTFOLIO (*Læringsmappen*)

A central component of the pedagogical framework is the *learning portfolio* (*læringsmappen*), which students build throughout the course. The portfolio serves both as a learning tool and as a formative assessment instrument. It is constructed step by step and consists of the following elements:

- **Why medication calculation is so important.** Students begin by articulating, in their own words, why medication calculation is a high-stakes competence. This includes reflecting on the consequences of errors, the ethical responsibility of the nurse, and the role of precision in patient safety [6], [7]. This section anchors the entire portfolio in professional accountability.
- **Problem solving and algorithmic thinking.** Students document how they approach medication calculation as a structured problem-solving process. This includes identifying the known quantities, selecting the appropriate formula, performing unit conversions, and verifying the result. Algorithmic thinking is emphasised as a general competence that transfers to other domains [5].
- **Medication calculation.** Students collect and solve a curated set of medication calculation problems covering tablets, liquids, weight-based dosages, IV drip rates, and unit conversions. Each problem is documented with the formula used, the step-by-step calculation, and a clinical comment on whether the result is plausible.
- **Prompt engineering and pitfalls.** Students document their own prompts to AI, classify them as “poor”, “adequate”, or “good”, and reflect on how prompt quality affected the usefulness of the AI response. They also identify common pitfalls in AI-generated medication calculations, such as unit errors, forgotten conversions, and hallucinated concentrations.
- **Some tasks with implementation of algorithmic thinking.** Students implement simple algorithms in Python (or pseudocode) for dose calculation and drip rate calculation. The purpose is not to become programmers, but to understand the logic of the calculation and to be able to verify AI-generated code.
- **Problem collection.** Students compile a personal collection of medication calculation problems, organised by category and difficulty. This collection becomes a study resource and a record of their progression.

- **Reflection and control questions.** Students answer a set of structured reflection and control questions designed to promote metacognition and self-assessment [3]. These questions ask students to explain concepts, compare methods, identify error sources, and evaluate their own competence.

A. Step-by-Step Construction of the Portfolio

The portfolio is built step by step over the course of the semester. Each step introduces new content and new reflection prompts.

1) *Step 1: Why Medication Calculation Is So Important:* Students write a short essay in which they reflect on the consequences of medication errors. They are asked to consider:

- The difference between a calculation error in a classroom and in a clinical setting.
- The ethical and legal responsibility of the nurse.
- The role of double-checking and verification in clinical practice [6].

This step establishes the *why* of the entire portfolio and motivates the subsequent work.

2) *Step 2: Problem Solving and Algorithmic Thinking:* Students are introduced to algorithmic thinking as a general problem-solving strategy [5]. They learn to:

- Decompose a problem into smaller steps.
- Identify the known and unknown quantities.
- Select and apply the appropriate formula.
- Verify the result using estimation and unit checks.

This step connects medication calculation to broader computational thinking competences.

3) *Step 3: Medication Calculation:* Students work through a curated set of medication calculation problems. Each problem is documented with:

- The original problem statement.
- The formula used.
- The step-by-step calculation.
- A clinical plausibility check.

The problems cover the following categories:

TABLE I
CATEGORIES OF MEDICATION CALCULATION PROBLEMS IN THE
LEARNING PORTFOLIO.

Category	Example
Tablet dosage	Toprol XL 50 mg ordered; 100 mg tablets available
Liquid dosage	Furosemide 40 mg ordered; 10 mg/ml available
Weight-based dosage	Paracetamol 15 mg/kg; patient weighs 72 kg
IV drip rate	250 ml over 180 min; drop factor 30 dr/ml
Unit conversion	64.8 lb to kg; 1 g to mg
Reconstitution	1 g powder to 20 mg/ml
Renal adjustment	50% dose reduction in renal failure

4) *Step 4: Prompt Engineering and Pitfalls:* Students document their own prompts to AI and reflect on prompt quality. They are introduced to the following classification:

“Give me the answer to this medication calculation.”

“I am calculating a dose of Furosemide for a patient. The order is 40 mg IV, and the available concentration is 10 mg/ml. Can you walk me through the calculation step by step, explain the formula, and check whether my result of 4 ml is correct?”

Students also identify common pitfalls in AI-generated medication calculations:

- **Unit errors:** AI may mix up mg and g, or ml and l.
- **Forgotten conversions:** AI may fail to convert lb to kg.
- **Hallucinated concentrations:** AI may invent a concentration that does not exist.
- **Overconfident errors:** AI may present a wrong answer with high confidence [20], [22].

5) *Step 5: Tasks with Implementation of Algorithmic Thinking:* Students implement simple algorithms in Python for dose calculation and drip rate calculation. The purpose is to understand the logic of the calculation and to be able to verify AI-generated code.

```
def dose_tablets(ordered_mg, available_mg):
    return ordered_mg / available_mg
```

```
def drip_rate(volume_ml, time_min, drop_factor):
    return (volume_ml / time_min) * drop_factor
```

Students are asked to test their code with the problems from Step 3 and to compare their results with AI-generated code.

6) *Step 6: Problem Collection:* Students compile a personal collection of medication calculation problems, organised by category and difficulty. This collection becomes a study resource and a record of their progression.

7) *Step 7: Reflection and Control Questions:* Students answer a set of structured reflection and control questions:

- **Safety:** How can you double-check that the AI has given you the correct answer? Which other sources should you consult?
- **Understanding vs. answer:** Did you get an explanation you understand, or just an answer you can copy?
- **Clinical judgment:** What information in a clinical scenario can the AI *not* take into account (e.g., renal function, interactions)?
- **Prompt quality:** How did the way you asked the question affect the quality of the answer?
- **Limits of AI:** In which types of medication calculations or situations would you *not* trust AI alone? Why?

B. Pedagogical Rationale for the Learning Portfolio

The learning portfolio is designed to promote deep learning and metacognition. It is grounded in the following pedagogical principles:

- **Constructive alignment:** The portfolio aligns learning outcomes, activities, and assessment [2].
- **Self-regulated learning:** Students plan, monitor, and evaluate their own learning [3].
- **Reflective practice:** Students articulate their reasoning and reflect on their errors [4].
- **Algorithmic thinking:** Students develop a general problem-solving competence [5].
- **Critical AI literacy:** Students learn to evaluate AI outputs critically [15], [17].

V. RESULTS

A. Observations from the Case Study

Key observations from the classroom implementation:

- Students using more elaborated prompts appeared to develop conceptual understanding more effectively than those who primarily copied answers.
- Many initially fell into Pitfall 1 (asking only for the answer). After prompt engineering and reflection, understanding improved.
- Flipped classroom + AI freed class time for conceptual discussion and clinical reasoning.
- Self-assessment helped students identify gaps and target practice.
- The learning portfolio provided a structured record of student progression and made reflection visible.
- The interdisciplinary cooperation between the two faculties enriched both the mathematical and the clinical dimensions of the teaching.

While these findings are based on qualitative observations and are not statistically generalisable, they provide indicative insights into how structured pedagogical design can influence student interaction with AI and support the development of more active and reflective learning strategies.

These observations can be interpreted through the lens of self-regulated learning. Students who used elaborated prompts demonstrated behaviours associated with planning, monitoring, and evaluation [3]. In contrast, students relying on minimal prompts exhibited surface learning strategies focused on answer retrieval rather than conceptual understanding [2].

The flipped classroom structure appears to amplify these differences, as it creates space for independent exploration prior to formal instruction. In this sense, AI acts as a multiplier of existing learning strategies: it supports deep learning when used reflectively, but reinforces superficial approaches when used uncritically.

B. Student Feedback on Metacognitive Benefits

Based on informal student feedback and reflective responses, several students reported that the reflection questions and self-assessment increased their awareness of their own

learning processes, not only in medication calculation but also in other subjects. One student said: “The self-assessment made me realise exactly what I need to work on, not just in maths but in pharmacology as well.” Another noted: “Reflecting on how I used AI helped me see that I was just copying answers before. Now I ask better questions in all my courses.” Several students explicitly stated that they transferred this metacognitive habit to other courses, where they began to ask themselves: “What do I actually understand here? What do I need to focus on?”

VI. DISCUSSION

The findings from the case study can be understood as an interaction between pedagogical design and AI affordances. The flipped classroom creates conditions for active engagement, while structured reflection transforms AI use into a metacognitive process. This supports the broader conclusion that the effectiveness of AI in education depends less on the technology itself and more on how it is embedded within instructional design [13], [14].

A. The Risk of Becoming Passive Users – Six Major Pitfalls

Despite its potential, uncritical AI use carries serious risks. The six pitfalls presented below are synthesised from both the literature and classroom observations.

Pitfall 1: AI as a fact machine – Students ask only for the final answer and copy it without understanding.

Prevention: Teach students to formulate prompts that request guidance and explanation, not just the final answer.

Pitfall 2: Uncritical acceptance of AI hallucinations – AI can produce confident but incorrect mathematics [20], [22].

Prevention: Deliberately show AI errors in class. Teach cross-checking using textbooks or formularies.

Pitfall 3: Passive learning and skill atrophy – Students conclude: “I don’t need to practice – AI will do it for me.”

Prevention: Use proctored assessments and reflective prompts. Mandate that certain core tasks be done without AI.

Pitfall 4: Over-trust and techno-authoritative conviction – Students treat AI as an unquestionable authority [17].

Prevention: Train students to identify errors. Use “adversarial examples” where the AI deliberately provides wrong answers for students to catch [21].

Pitfall 5: Erosion of foundational skills (brittle knowledge) – Skipping basic skill-building leads to a “brittle” knowledge base [19].

Prevention: Mandate certain core tasks be completed without AI. Design assessments that specifically test foundational skills.

Pitfall 6: Loss of professional identity and agency – Over-reliance erodes students’ sense of themselves as capable practitioners [18].

Prevention: Normalise struggle as part of learning. Use reflective journals where students document their problem-solving process.

B. Implications for Teaching

The results show that structured reflection and self-assessment are essential for preventing these six pitfalls. When students regularly articulate their understanding, identify gaps, and evaluate progress, they build metacognitive skills that transfer beyond medication calculation. Self-assessment moves students from passive receipt of AI-generated answers to active monitoring of their own competence.

Student feedback confirms that these practices have increased awareness of learning processes across subjects. We therefore recommend that any AI integration in medication calculation education be accompanied by systematic opportunities for reflection and self-assessment. These practices restore student agency and professional identity, countering over-dependence and passive consumption.

The interdisciplinary cooperation between the Faculty of Technology, Environmental and Social Sciences and the Faculty of Health and Social Sciences has been essential in achieving this. The mathematical and algorithmic perspectives from the technology faculty ensure that the calculation methods are rigorous and verifiable, while the clinical perspectives from the health faculty ensure that the pedagogical design remains anchored in patient safety and professional responsibility.

C. Limitations and Future Work

Our findings are based on a single case study at one institution. Future work will evaluate a multi-agent AI tutoring platform with adaptive feedback and a task bank for personalised problem generation [12]. We also plan to investigate long-term effects on professional identity and self-efficacy.

VII. CONCLUSION

Artificial intelligence is not a temporary trend but a permanent fixture in the educational landscape. Attempting to ban or restrict student access to AI is both impractical and pedagogically misguided. Instead, we must teach students to use AI as a *training partner*, a tool that supports, but never replaces, their own thinking.

From the Student's Perspective

Students must move from passive consumers of AI-generated answers to active, self-regulated learners. This means:

- **Attempting first:** Always trying to solve a problem on their own before consulting AI.
- **Prompting with purpose:** Asking for explanations, verification, or alternative approaches, not just final answers.
- **Reflecting on outcomes:** Using checklists and self-assessment questions to evaluate what they have learned and what remains unclear.
- **Practicing without AI:** Regularly testing themselves in AI-free environments to ensure genuine skill development.

From the Teacher's Perspective

The teacher's role evolves from being the sole source of answers to a designer of learning experiences that require uniquely human cognitive work. Key responsibilities include:

- **Designing reflective tasks:** Assignments that ask students to compare, critique, and justify AI outputs, not just compute.
- **Teaching AI literacy explicitly:** Demonstrating good prompting, pointing out common AI errors, and discussing ethical use.
- **Creating formative self-assessment tools:** Checklists, rubrics, and reflection prompts that help students monitor their own learning.
- **Maintaining proctored assessments:** Ensuring that some portion of evaluation remains AI-free to verify individual competency.

The Pedagogical Core

Our case study with medication calculation provides indications of improved conceptual understanding and increased problem-solving confidence. The success of this approach hinges on three pedagogical pillars:

- 1) **Good prompting:** teaching students how to ask questions that elicit explanation, not just answers.
- 2) **Reflection and self-assessment:** requiring students to articulate what they learned, what they still struggle with, and how they plan to improve.
- 3) **Formative tasks with critical evaluation:** designing assignments where students must compare, verify, and critique AI-generated solutions.

The Ultimate Goal

The purpose of this pedagogical approach is not to restrict AI use, but to facilitate good use of AI that contributes to deep learning. We want students to understand medication calculation deeply enough to use AI as a partner in exploration, a checker of routine work, and a generator of alternative perspectives. At the same time, AI can be used in multiple ways to increase learning outcomes and streamline guidance.

Reflection, self-assessment, well-designed control questions, and formative tasks that require students to apply, compare, and justify – these are the pedagogical keys to achieving that goal. By embedding these practices into everyday teaching, we transform medication calculation education into a journey of human intellectual growth, preparing students not just for exams, but for a lifetime of thoughtful, responsible problem-solving in an AI-augmented clinical world.

The cooperation between the Faculty of Technology, Environmental and Social Sciences and the Faculty of Health and Social Sciences at HVL demonstrates that interdisciplinary collaboration is not merely desirable but necessary when addressing high-stakes educational challenges at the intersection of technology and health.

TABLE II
KEY RECOMMENDATIONS FOR INTEGRATING AI IN MEDICATION
CALCULATION EDUCATION.

1. Teach AI literacy: good prompting (ask for explanation, not just answers), error detection, ethical use.
2. Require students to attempt problems first, then use AI for verification and reflection.
3. Use AI for formative feedback, but maintain proctored, AI-free summative assessments.
4. Leverage AI for teaching: generate practice problems, analyse misconceptions, differentiate tasks.
5. Create clear institutional policies on permitted AI use and documentation.
6. Address the six pitfalls explicitly in teaching.
7. Integrate structured self-assessment, reflection questions, and control questions into every AI-augmented activity.
8. Build a learning portfolio that documents the student's progression step by step.

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